Boolean Fourier Analysis and the Walsh–Hadamard Transform

An Expository Guide

From First Principles to Spectral Methods
for Binary Neural Networks

Companion to the thesis proposal:
Spectral Methods for Binary Neural Networks

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This document provides a self-contained exposition of the mathematical background needed to read the thesis proposal on spectral methods for binary neural networks. It assumes familiarity with undergraduate linear algebra, probability, and calculus, but develops all specialized material—Boolean Fourier analysis, the Walsh–Hadamard transform, hypercontractivity, influence theory, and their connections to coding theory and neural networks—from scratch, with full proofs and references.

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Chapter 1

Prerequisites

We collect the facts from linear algebra and probability that are used throughout this guide. The reader who is comfortable with inner product spaces, the Kronecker product, Markov's inequality, and the central limit theorem may safely skip to Chapter 2.

Standard references: Axler [2024] or Horn and Johnson [2012] for linear algebra; Durrett [2019] or Feller [1971] for probability.

1.1 Inner Product Spaces

Definition 1.1.1 (Inner product)**.** Let V be a real vector space. An *inner product* on V is a function $\langle \cdot, \cdot \rangle \colon V \times V \to \mathbb{R}$ satisfying, for all $u, v, w \in V$ and $\alpha \in \mathbb{R}$:

(i) *Symmetry:* $\langle u, v \rangle = \langle v, u \rangle$.

(ii) *Linearity in the first argument:* $\langle \alpha u + v, w \rangle = \alpha \langle u, w \rangle + \langle v, w \rangle$.

(iii) *Positive definiteness:* $\langle v, v \rangle \geq 0$, with equality if and only if $v = 0$.

The pair $(V, \langle \cdot, \cdot \rangle)$ is called an *inner product space*.

The inner product induces a norm $\|v\| = \sqrt{\langle v, v \rangle}$. The key structural fact is:

Theorem 1.1.2 (Cauchy–Schwarz inequality)**.** *For all u, v in an inner product space, $|\langle u, v \rangle| \leq \|u\| \, \|v\|$, with equality if and only if u and v are linearly dependent.*

Definition 1.1.3 (Orthonormal basis)**.** A set $\{e_1, \ldots, e_n\}$ in an inner product space V is *orthonormal* if $\langle e_i, e_j \rangle = \delta_{ij}$ (the Kronecker delta). If additionally $\operatorname{span}\{e_1, \ldots, e_n\} = V$, it is an *orthonormal basis* (ONB).

The central fact about orthonormal bases is:

Proposition 1.1.4 (Fourier expansion in a finite-dimensional inner product space)**.** *If* $\{e_1, \ldots, e_N\}$ *is an ONB for* $(V, \langle \cdot, \cdot \rangle)$*, then every* $v \in V$ *has a unique expansion*

$$v = \sum_{i=1}^{N} \langle v, e_i \rangle \, e_i,$$

and Parseval's identity *holds:* $\|v\|^2 = \sum_{i=1}^{N} \langle v, e_i \rangle^2$.

Proof. Write $v = \sum_i c_i e_i$. Taking the inner product with e_j gives $\langle v, e_j \rangle = c_j$ by orthonormality. Then $\|v\|^2 = \langle v, v \rangle = \sum_i c_i^2$. □

This is exactly the abstract template for the Boolean Fourier expansion in Chapter 2: the vector space will be the space of all real-valued functions on $\{-1, +1\}^n$, the inner product will be the expectation inner product, and the ONB will be the parity functions.

1.2 The Kronecker Product

Definition 1.2.1 (Kronecker product)**.** Let A be an $m \times n$ matrix and B a $p \times q$ matrix. Their *Kronecker product* $A \otimes B$ is the $mp \times nq$ block matrix

$$A \otimes B = \begin{pmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}B & a_{m2}B & \cdots & a_{mn}B \end{pmatrix}.$$

Key properties (all straightforward to verify):

(i) $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$, when the products AC and BD are defined.

(ii) $(A \otimes B)^\top = A^\top \otimes B^\top$.

(iii) If A and B are orthogonal, so is $A \otimes B$.

(iv) $\operatorname{tr}(A \otimes B) = \operatorname{tr}(A) \operatorname{tr}(B)$.

Property (i) is the *mixed-product rule*; it is the algebraic engine behind the recursive structure of the Hadamard matrix (Chapter 5).

Example 1.2.2. The Hadamard matrix $H_1 = \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$ satisfies

$$H_1 \otimes H_1 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} = H_2.$$

See Section 5.1 for the general construction.

1.3 Probability Essentials

Throughout this guide, probability spaces are finite (typically uniform on $\{-1,+1\}^n$), so measure-theoretic subtleties do not arise.

Definition 1.3.1 (Expectation and variance)**.** For a random variable X on a finite probability space (Ω, p):

$$\mathbb{E}[X] = \sum_{\omega\in\Omega} p(\omega)\, X(\omega), \qquad \mathrm{Var}[X] = \mathbb{E}[(X-\mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

Theorem 1.3.2 (Markov's inequality)**.** *If $X \geq 0$ and $a > 0$, then $\mathbb{P}[X \geq a] \leq \mathbb{E}[X]/a$.*

Proof. $\mathbb{E}[X] = \sum_\omega p(\omega)X(\omega) \geq \sum_{\omega: X(\omega)\geq a} p(\omega)\cdot a = a\cdot \mathbb{P}[X\geq a]$. □

Corollary 1.3.3 (Chebyshev's inequality)**.** $\mathbb{P}[|X-\mathbb{E}[X]| \geq t] \leq \mathrm{Var}[X]/t^2$.

Proof. Apply Markov to $(X-\mathbb{E}[X])^2$ with $a = t^2$. □

Markov's inequality is used directly in the thesis to prove the degree-truncation approximation bound (Proposition 3.1 of the thesis).

1.4 The Central Limit Theorem and Berry–Esséen

Theorem 1.4.1 (Central Limit Theorem)**.** *Let $X_1, X_2, \ldots$ be independent, identically distributed random variables with $\mathbb{E}[X_i] = 0$ and $\mathrm{Var}[X_i] = \sigma^2 > 0$. Then*

$$\frac{X_1 + \cdots + X_n}{\sigma\sqrt{n}} \xrightarrow{d} \mathcal{N}(0,1) \quad \text{as } n \to \infty.$$

The CLT tells us that $S_n/(\sigma\sqrt{n})$ is approximately $\mathcal{N}(0,1)$ for large n, but says nothing about the rate of convergence. The Berry–Esséen theorem quantifies this.

Theorem 1.4.2 (Berry–Esséen; Berry, 1941, Esséen, 1942)**.** *Under the hypotheses of Theorem 1.4.1, with additionally $\mathbb{E}[|X_i|^3] = \rho < \infty$, we have*

$$\sup_{t\in\mathbb{R}} \left| \mathbb{P}\left[\frac{S_n}{\sigma\sqrt{n}} \leq t\right] - \Phi(t) \right| \leq \frac{C\rho}{\sigma^3\sqrt{n}},$$

where Φ is the standard normal CDF and C is a universal constant (the best known value is $C < 0.4748$).

Why this matters for the thesis. In the thesis, binary neural network layers compute sums of the form $y_j = \sum_{i=1}^{d} w_{ji} x_i$ where $w_{ji}, x_i \in \{-1, +1\}$. When x is uniform on $\{-1, +1\}^d$, the summands $w_{ji} x_i$ are independent ± 1 random variables (regardless of the fixed signs w_{ji}), so $\mathbb{E}[w_{ji} x_i] = 0$, $\mathrm{Var}[w_{ji} x_i] = 1$, and $\mathbb{E}[|w_{ji} x_i|^3] = 1$. By Berry–Esséen, $y_j/\sqrt{d}$ is within $O(1/\sqrt{d})$ of $\mathcal{N}(0, 1)$ in CDF. At $d = 1024$, this is about 0.015—excellent approximation. This justifies the $1/\sqrt{d}$ normalization used throughout the thesis as a replacement for BitNet's learned scaling factors.

1.5 L^p Norms

For a function f on a probability space (Ω, p) and $p \geq 1$:

Definition 1.5.1 (L^p norm)**.**

$$\|f\|_p = (\mathbb{E}[|f|^p])^{1/p} = \left(\sum_{\omega \in \Omega} p(\omega) \, |f(\omega)|^p \right)^{1/p}.$$

For $p = \infty$: $\|f\|_\infty = \max_\omega |f(\omega)|$.

We always have $\|f\|_p \leq \|f\|_q$ when $p \leq q$ (by Jensen's inequality; the direction may seem backwards, but recall we are on a probability space). The *hypercontractivity* results in Chapter 4 are striking precisely because they give a reverse inequality $\|T_\rho f\|_q \leq \|f\|_p$ for $q > p$, at the cost of applying the noise operator T_ρ.

Remark 1.5.2 (Notation). Throughout this guide, when we write $\mathbb{E}_x[\cdot]$ for functions on $\{-1, +1\}^n$, we mean the expectation under the uniform measure: each $x \in \{-1, +1\}^n$ has probability 2^{-n}. So $\|f\|_2^2 = \mathbb{E}_x[f(x)^2] = 2^{-n} \sum_{x \in \{-1,+1\}^n} f(x)^2$.

Chapter 2

Fourier Analysis on the Boolean Hypercube

This chapter develops the Fourier analysis of functions on $\{-1,+1\}^n$ from scratch. The treatment follows O'Donnell [2014], Chapters 1–2, which is the standard reference. A shorter introduction is de Wolf [2008].

2.1 The Boolean Hypercube

Definition 2.1.1 (Boolean hypercube)**.** The *Boolean hypercube* of dimension n is the set

$$\Omega_n = \{-1,+1\}^n = \{(x_1,\ldots,x_n) : x_i \in \{-1,+1\}\}.$$

It has $|\Omega_n| = 2^n$ elements. We equip it with the *uniform probability measure*: each $x \in \Omega_n$ has probability 2^{-n}.

Remark 2.1.2 (Why $\{-1,+1\}$ and not $\{0,1\}$?). Both conventions appear in the literature. The $\{-1,+1\}$ convention is standard in Boolean Fourier analysis because it makes the algebra cleaner:

- Multiplication is the natural group operation ($x_i \cdot x_j \in \{-1,+1\}$), whereas $\{0,1\}$ would require XOR (addition mod 2).
- The Fourier basis functions (parity functions, below) take values in $\{-1,+1\}$, matching the "alphabet" of Boolean functions.
- The connection to the Walsh–Hadamard transform (Chapter 5) is immediate.

To convert: if $b \in \{0,1\}$, set $x = 1 - 2b = (-1)^b$.

Definition 2.1.3 (Boolean and pseudo-Boolean functions)**.** A *Boolean function* is a map $f\colon \{-1,+1\}^n \to \{-1,+1\}$. A *pseudo-Boolean function* is a map $f\colon \{-1,+1\}^n \to \mathbb{R}$.

The set of all pseudo-Boolean functions $f\colon \{-1,+1\}^n \to \mathbb{R}$ forms a real vector space of dimension 2^n (one degree of freedom per point in Ω_n). We equip it with the *expectation inner product*:

Definition 2.1.4 (Inner product on functions)**.** For $f, g\colon \{-1,+1\}^n \to \mathbb{R}$:

$$\langle f, g\rangle = \mathbb{E}_x[f(x)\,g(x)] = \frac{1}{2^n} \sum_{x\in\{-1,+1\}^n} f(x)\,g(x). \tag{2.1}$$

The associated norm is $\|f\|_2 = \sqrt{\langle f, f\rangle} = \sqrt{\mathbb{E}_x[f(x)^2]}$.

Example 2.1.5. If $f\colon \{-1,+1\}^n \to \{-1,+1\}$ is Boolean, then $f(x)^2 = 1$ for all x, so $\|f\|_2^2 = 1$. Every Boolean function is a unit vector in this inner product space.

2.2 The Parity Functions

We now construct an orthonormal basis for the space of pseudo-Boolean functions. The key objects are the *parity functions* (also called *Walsh functions* or *characters*).

Definition 2.2.1 (Parity function)**.** For each subset $S \subseteq [n] = \{1, 2, \ldots, n\}$, define

$$\chi_S(x) = \prod_{i\in S} x_i. \tag{2.2}$$

By convention, $\chi_\emptyset(x) = 1$ for all x (the empty product).

Since each $x_i \in \{-1,+1\}$, each $\chi_S(x)$ is a product of ± 1 values, hence $\chi_S(x) \in \{-1,+1\}$. The parity functions are themselves Boolean functions. This is a distinctive feature of the Boolean Fourier basis: the basis elements live in the same alphabet as the functions being analyzed. (Compare with classical Fourier analysis, where the basis $\{e^{2\pi i k x}\}$ is complex-valued even when analyzing real functions.)

Example 2.2.2. For $n = 3$:

$$\begin{aligned}
\chi_\emptyset(x) &= 1,\\
\chi_{\{1\}}(x) &= x_1, \quad \chi_{\{2\}}(x) = x_2, \quad \chi_{\{3\}}(x) = x_3,\\
\chi_{\{1,2\}}(x) &= x_1x_2, \quad \chi_{\{1,3\}}(x) = x_1x_3, \quad \chi_{\{2,3\}}(x) = x_2x_3,\\
\chi_{\{1,2,3\}}(x) &= x_1x_2x_3.
\end{aligned}$$

There are $2^3 = 8$ such functions, one for each subset of $[3]$.

2.3 Orthonormality

Theorem 2.3.1 (Orthonormality of parity functions)**.** *For any $S, T \subseteq [n]$:*

$$\langle \chi_S, \chi_T \rangle = \delta_{ST} = \begin{cases} 1 & \text{if } S = T, \\ 0 & \text{if } S \neq T. \end{cases} \tag{2.3}$$

Proof. We compute:

$$\langle \chi_S, \chi_T \rangle = \mathbb{E}_x \left[\prod_{i \in S} x_i \cdot \prod_{j \in T} x_j \right] = \mathbb{E}_x \left[\prod_{i \in S \triangle T} x_i \right],$$

where $S \triangle T = (S \setminus T) \cup (T \setminus S)$ is the symmetric difference (we used $x_i^2 = 1$ to cancel shared indices).

If $S = T$, then $S \triangle T = \emptyset$, the product is 1, and the expectation is 1.

If $S \neq T$, then $S \triangle T \neq \emptyset$; pick any $k \in S \triangle T$. Since x_k is independent of $\{x_i : i \neq k\}$ and takes ± 1 with equal probability:

$$\mathbb{E}_x \left[\prod_{i \in S \triangle T} x_i \right] = \mathbb{E}_{x_k}[x_k] \cdot \mathbb{E}_{x_{-k}} \left[\prod_{\substack{i \in S \triangle T \\ i \neq k}} x_i \right] = 0 \cdot (\text{something}) = 0.$$

(Here x_{-k} denotes the coordinates other than k.) □

Since there are 2^n parity functions and the space of pseudo-Boolean functions has dimension 2^n, orthonormality implies:

Corollary 2.3.2. *$\{\chi_S\}_{S \subseteq [n]}$ is an orthonormal basis for the space of all functions $f \colon \{-1, +1\}^n \to \mathbb{R}$ under the inner product $\langle f, g \rangle = \mathbb{E}_x[f(x)g(x)]$.*

2.4 The Fourier Expansion

By Proposition 1.1.4 (general Fourier expansion) and Corollary 2.3.2 (the parity functions form an ONB), we immediately obtain:

Theorem 2.4.1 (Fourier expansion on $\{-1, +1\}^n$)**.** *Every function $f \colon \{-1, +1\}^n \to \mathbb{R}$ has a unique expansion*

$$f(x) = \sum_{S \subseteq [n]} \widehat{f}(S)\, \chi_S(x), \tag{2.4}$$

where the Fourier coefficients *(or* Walsh–Fourier coefficients*) are*

$$\widehat{f}(S) = \langle f, \chi_S \rangle = \mathbb{E}_x\left[f(x)\prod_{i\in S} x_i\right] = \frac{1}{2^n}\sum_{x\in\{-1,+1\}^n} f(x)\prod_{i\in S} x_i. \tag{2.5}$$

The expansion (2.4) is a *multilinear polynomial*: each variable x_i appears with degree at most 1 (because $x_i^2 = 1$ for $x_i \in \{-1, +1\}$, so any higher power reduces). This is the unique multilinear representation of f.

Definition 2.4.2 (Degree)**.** The *degree* of the monomial χ_S is $|S|$. We say f has *Fourier degree at most* d if $\widehat{f}(S) = 0$ for all S with $|S| > d$.

2.5 Parseval's Identity

Applying Proposition 1.1.4 (Parseval in the abstract setting) with the parity ONB:

Theorem 2.5.1 (Parseval's identity)**.** *For any* $f\colon \{-1, +1\}^n \to \mathbb{R}$*:*

$$\|f\|_2^2 = \mathbb{E}_x[f(x)^2] = \sum_{S\subseteq[n]} \widehat{f}(S)^2. \tag{2.6}$$

For Boolean functions ($f\colon \{-1, +1\}^n \to \{-1, +1\}$):

$$\sum_{S\subseteq[n]} \widehat{f}(S)^2 = 1. \tag{2.7}$$

This is the *energy constraint*: the Fourier coefficients of a Boolean function lie on the unit sphere in $\mathbb{R}^{2^n}$. The "spectral energy" $\widehat{f}(S)^2$ at each subset S can be thought of as the fraction of the function's total variance explained by that frequency component.

Remark 2.5.2 (Relevance to the thesis). The thesis uses Parseval's identity in a crucial way: it implies that the Fourier coefficients of any Boolean function are automatically normalized ($\sum_S \widehat{f}(S)^2 = 1$). This means the spectral parameterization of binary weight matrices comes with a built-in energy budget, replacing the learned per-layer scaling factors α used in BitNet.

2.6 Worked Examples

We compute the Fourier expansion for several important Boolean functions. These examples build intuition for "spectral concentration"—the phenomenon that many natural Boolean functions have most of their Fourier mass on low-degree coefficients.

2.6.1 Dictator Functions

Example 2.6.1 (Dictator)**.** The i-th *dictator function* is $f(x) = x_i$. Its Fourier expansion is trivially $f = \chi_{\{i\}}$, so

$$\widehat{f}(S) = \begin{cases} 1 & \text{if } S = \{i\}, \\ 0 & \text{otherwise.} \end{cases}$$

All spectral energy is at degree 1. Parseval: $1^2 = 1$. ✓

Dictator functions are the extreme case of spectral concentration: all energy is on a single degree-1 coefficient.

2.6.2 The AND Function

Example 2.6.2 (AND on 2 bits)**.** Define $\mathrm{AND}(x_1, x_2) = +1$ iff $x_1 = x_2 = +1$, and -1 otherwise. The truth table:

x_1	x_2	AND
-1	-1	-1
-1	$+1$	-1
$+1$	-1	-1
$+1$	$+1$	$+1$

Computing coefficients:

$$\widehat{f}(\emptyset) = \mathbb{E}[f] = \tfrac{1}{4}(-1 - 1 - 1 + 1) = -\tfrac{1}{2},$$
$$\widehat{f}(\{1\}) = \mathbb{E}[f \cdot x_1] = \tfrac{1}{4}(1 + 1 - 1 - 1) = 0,$$

Wait—let us redo this more carefully. We have $\mathbb{E}[f \cdot x_1] = \frac{1}{4}[(-1)(-1) + (-1)(-1) + (-1)(1) + (1)(1)] = \frac{1}{4}[1 + 1 - 1 + 1] = \frac{1}{2}$.

Similarly $\widehat{f}(\{2\}) = \frac{1}{2}$ and $\widehat{f}(\{1, 2\}) = \mathbb{E}[f \cdot x_1 x_2] = \frac{1}{4}[(-1)(1) + (-1)(-1) + (-1)(-1) + (1)(1)] = \frac{1}{4}[-1 + 1 + 1 + 1] = \frac{1}{2}$.

So: $\mathrm{AND}(x_1, x_2) = -\frac{1}{2} + \frac{1}{2}x_1 + \frac{1}{2}x_2 + \frac{1}{2}x_1 x_2$.

Parseval: $\frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = 1$. ✓

The spectral energy is spread evenly across degrees 0, 1, 1, 2.

2.6.3 The Majority Function

Example 2.6.3 (Majority on 3 bits)**.** $\mathrm{Maj}_3(x) = \mathrm{sgn}(x_1 + x_2 + x_3)$: output $+1$ if at least 2 of 3 inputs are $+1$.

Truth table and Fourier computation:

x_1	x_2	x_3	Maj_3
-1	-1	-1	-1
-1	-1	$+1$	-1
-1	$+1$	-1	-1
-1	$+1$	$+1$	$+1$
$+1$	-1	-1	-1
$+1$	-1	$+1$	$+1$
$+1$	$+1$	-1	$+1$
$+1$	$+1$	$+1$	$+1$

The Fourier coefficients are:

$$\begin{aligned}
\widehat{f}(\emptyset) &= 0 \quad \text{(balanced: 4 outputs } +1\text{, 4 outputs } -1\text{)}, \\
\widehat{f}(\{1\}) = \widehat{f}(\{2\}) = \widehat{f}(\{3\}) &= \tfrac{1}{2}, \\
\widehat{f}(\{1,2\}) = \widehat{f}(\{1,3\}) = \widehat{f}(\{2,3\}) &= 0, \\
\widehat{f}(\{1,2,3\}) &= -\tfrac{1}{2}.
\end{aligned}$$

Hence: $\mathrm{Maj}_3(x) = \frac{1}{2}(x_1 + x_2 + x_3) - \frac{1}{2}x_1x_2x_3$.
Parseval: $3 \cdot \frac{1}{4} + \frac{1}{4} = 1$. ✓
The spectral energy breakdown by degree:

Degree	Energy	Fraction
0	0	0%
1	3/4	75%
2	0	0%
3	1/4	25%

Maj_3 is *spectrally concentrated*: 75% of its energy is at degree 1. This is a general phenomenon: the majority function on n bits has most of its spectral energy on degree-1 coefficients, with the concentration improving as n grows [O'Donnell, 2014, Chapter 5].

2.6.4 The Parity Function (Anti-Concentration)

Example 2.6.4 (Parity)**.** $f(x) = x_1x_2 \cdots x_n = \chi_{[n]}(x)$.

This function has $\widehat{f}([n]) = 1$ and all other coefficients zero. All energy is at the maximum degree n. The parity function is the *opposite* of spectrally concentrated: it is maximally complex in the Fourier sense. It is also the hardest Boolean function to learn from random examples [Linial et al., 1993].

2.7 Degree-Level Energy Decomposition

Parseval allows us to decompose the "energy" of a Boolean function by degree:

Definition 2.7.1 (Spectral energy at degree k)**.** For $f\colon \{-1,+1\}^n \to \mathbb{R}$ and $0 \le k \le n$:

$$W^k[f] = \sum_{\substack{S\subseteq[n]\\|S|=k}} \widehat{f}(S)^2.$$

The *spectral energy profile* is the vector $(W^0[f], W^1[f], \ldots, W^n[f])$.

By Parseval: $\sum_{k=0}^{n} W^k[f] = \|f\|_2^2$. For Boolean f, this sum is 1, so the energy profile is a probability distribution on degrees $\{0, 1, \ldots, n\}$.

For a *random* Boolean function (chosen uniformly at random from all 2^{2^n} Boolean functions on n bits), the expected energy at degree k is $\binom{n}{k}/2^n$. This follows a Binomial$(n, 1/2)$ distribution, peaked at $k = n/2$. So a random Boolean function has its energy concentrated around degree $n/2$.

The thesis's central hypothesis is that *trained* binary weight matrices, viewed as Boolean functions, have their energy concentrated at much lower degrees—far below the $n/2$ baseline of random functions.

2.8 Low-Degree Truncation

Definition 2.8.1 (Degree-k truncation)**.** For $f\colon \{-1,+1\}^n \to \mathbb{R}$:

$$f^{\le k}(x) = \sum_{\substack{S\subseteq[n]\\|S|\le k}} \widehat{f}(S)\,\chi_S(x).$$

Similarly, $f^{>k} = f - f^{\le k}$.

The truncation $f^{\le k}$ is the best degree-k approximation to f in the L^2 norm (this follows from orthogonality: the "tail" $f^{>k}$ is orthogonal to all degree-$\le k$ functions). The approximation error is:

$$\left\|f - f^{\le k}\right\|_2^2 = \left\|f^{>k}\right\|_2^2 = \sum_{|S|>k} \widehat{f}(S)^2. \tag{2.8}$$

For Boolean functions, if we want to *round* $f^{\le k}$ back to $\{-1,+1\}$ via the sign function, the fraction of inputs where the rounded version disagrees with f is bounded by:

Proposition 2.8.2 (Sign-rounding error; thesis Proposition 3.1)**.** *Let* $f\colon \{-1,+1\}^n \to \{-1,+1\}$ *be Boolean and* $\tilde{f} = \operatorname{sgn}(f^{\leq k})$*. Then*

$$\mathbb{P}_x[\tilde{f}(x) \neq f(x)] \leq \sum_{|S|>k} \widehat{f}(S)^2.$$

Proof. If $\tilde{f}(x) \neq f(x)$, then $f(x) \cdot f^{\leq k}(x) \leq 0$. Write $f^{\leq k}(x) = f(x) - f^{>k}(x)$, so $f(x) \cdot f^{\leq k}(x) = 1 - f(x) \cdot f^{>k}(x)$. Thus the disagreement requires $f(x) \cdot f^{>k}(x) \geq 1$, which implies $|f^{>k}(x)| \geq 1$. By Markov's inequality (Theorem 1.3.2) applied to $(f^{>k})^2$:

$$\mathbb{P}[|f^{>k}(x)| \geq 1] \leq \mathbb{E}[(f^{>k}(x))^2] = \sum_{|S|>k} \widehat{f}(S)^2. \qquad \square$$

This bound is tight when $f^{>k}$ is small everywhere it is nonzero (as happens for spectrally concentrated functions), and loose when $f^{>k}$ takes large values on a small set.

Example 2.8.3 (Truncation of Maj_3)**.** For Maj_3, the tail energy above degree 1 is $\widehat{f}(\{1,2,3\})^2 = 1/4$, so the bound gives $\mathbb{P}[\operatorname{sgn}(f^{\leq 1}) \neq f] \leq 1/4$. In fact, $f^{\leq 1}(x) = \frac{1}{2}(x_1 + x_2 + x_3)$, which has the same sign as Maj_3 on all 8 inputs (since $|x_1+x_2+x_3| \geq 1$ always), so the actual error is $0/8 = 0$. The bound is not tight here because $f^{>1}(x) = -\frac{1}{2}x_1x_2x_3$ never has magnitude exceeding $\frac{1}{2} < 1$.

2.9 Plancherel and the Convolution Theorem

Two additional Fourier identities that are used in the thesis:

Theorem 2.9.1 (Plancherel's identity)**.** *For* $f, g\colon \{-1,+1\}^n \to \mathbb{R}$*:*

$$\langle f, g\rangle = \sum_{S\subseteq[n]} \widehat{f}(S)\,\widehat{g}(S).$$

Proof. Expand f and g in the Fourier basis and use orthonormality. $\square$

Definition 2.9.2 (Dyadic convolution)**.** For $f, g\colon \{-1,+1\}^n \to \mathbb{R}$, the *dyadic convolution* is

$$(f * g)(x) = \mathbb{E}_y[f(y)\, g(x \cdot y)],$$

where $x \cdot y = (x_1y_1, \ldots, x_ny_n)$ is coordinatewise multiplication.

Theorem 2.9.3 (Convolution theorem)**.** $\widehat{f * g}(S) = \widehat{f}(S) \cdot \widehat{g}(S)$.

Proof.

$$\begin{aligned}\widehat{f * g}(S) &= \mathbb{E}_x[(f * g)(x)\,\chi_S(x)] = \mathbb{E}_x[\mathbb{E}_y[f(y)\, g(x \cdot y)]\,\chi_S(x)] \\ &= \mathbb{E}_y[f(y)\,\mathbb{E}_x[g(x \cdot y)\,\chi_S(x)]]\,.\end{aligned}$$

Substituting $z = x \cdot y$ (which is uniform when x is uniform, since y is fixed and coordinatewise multiplication by $y \in \{-1,+1\}^n$ is a bijection on $\{-1,+1\}^n$):

$$\mathbb{E}_x[g(x \cdot y)\,\chi_S(x)] = \mathbb{E}_z[g(z)\,\chi_S(z \cdot y)] = \mathbb{E}_z[g(z)\,\chi_S(z)\,\chi_S(y)] = \widehat{g}(S)\,\chi_S(y),$$

using $\chi_S(z \cdot y) = \chi_S(z)\,\chi_S(y)$ (since χ_S is multiplicative: $\prod_{i \in S}(z_i y_i) = \prod_{i \in S} z_i \cdot \prod_{i \in S} y_i$).

Therefore: $\widehat{f * g}(S) = \widehat{g}(S)\,\mathbb{E}_y[f(y)\,\chi_S(y)] = \widehat{g}(S)\,\widehat{f}(S)$. $\square$

Chapter 3

Influence and Noise Sensitivity

This chapter introduces three related concepts—influence, noise sensitivity, and the noise operator—that quantify how "sensitive" a Boolean function is to perturbations of its input. Each has a clean spectral characterization, linking combinatorial/probabilistic notions to the Fourier coefficients. The main reference is O'Donnell [2014, Chapters 2–3].

3.1 Influence of Variables

Informally, the *influence* of variable i measures how often flipping x_i changes the output of f.

Definition 3.1.1 (Influence)**.** For $f\colon \{-1,+1\}^n \to \{-1,+1\}$ and $i \in [n]$, the *influence of variable* i is

$$\mathrm{Inf}_i[f] = \mathbb{P}_x[f(x) \neq f(x^{\oplus i})], \tag{3.1}$$

where $x^{\oplus i}$ denotes x with the i-th coordinate flipped: $(x_1, \ldots, -x_i, \ldots, x_n)$.

Theorem 3.1.2 (Spectral formula for influence)**.**

$$\mathrm{Inf}_i[f] = \sum_{S \ni i} \widehat{f}(S)^2. \tag{3.2}$$

Proof. Define $D_i f(x) = \frac{1}{2}(f(x) - f(x^{\oplus i}))$, the *discrete derivative* in direction i. Then $f(x) \neq f(x^{\oplus i})$ iff $|D_i f(x)| = 1$ (since f takes values ± 1, the difference is 0 or ± 2). Thus:

$$\mathrm{Inf}_i[f] = \mathbb{E}_x[(D_i f(x))^2].$$

Now compute the Fourier expansion of $D_i f$. Since $\chi_S(x^{\oplus i}) = (-1)^{[i \in S]}\chi_S(x)$ (flipping x_i negates χ_S iff $i \in S$):

$$D_i f(x) = \tfrac{1}{2} \sum_S \widehat{f}(S)\, \chi_S(x)\, (1 - (-1)^{[i \in S]}) = \sum_{S \ni i} \widehat{f}(S)\, \chi_S(x).$$

By Parseval: $\mathbb{E}[(D_i f)^2] = \sum_{S \ni i} \widehat{f}(S)^2$. □

Definition 3.1.3 (Total influence)**.**

$$\mathbf{I}[f] = \sum_{i=1}^{n} \mathrm{Inf}_i[f] = \sum_{S \subseteq [n]} |S| \cdot \widehat{f}(S)^2. \tag{3.3}$$

The total influence is the expected "degree" under the spectral distribution $\{\widehat{f}(S)^2\}_{S \subseteq [n]}$. Functions with low-degree spectral concentration necessarily have low total influence, and conversely.

Example 3.1.4 (Influences of example functions)**.**

- **Dictator** $f(x) = x_1$**:** $\mathrm{Inf}_1[f] = 1$, $\mathrm{Inf}_i[f] = 0$ for $i > 1$. Total: $\mathbf{I}[f] = 1$.
- **Parity** $f(x) = x_1 \cdots x_n$**:** $\mathrm{Inf}_i[f] = 1$ for all i. Total: $\mathbf{I}[f] = n$.
- $\mathbf{Maj}_3$**:** $\mathrm{Inf}_i[f] = \widehat{f}(\{i\})^2 + \widehat{f}(\{1,2,3\})^2 = 1/4 + 1/4 = 1/2$ for each i. Total: $\mathbf{I}[f] = 3/2$.

The total influence is bounded: $\mathbf{I}[f] \leq n$ for any Boolean f (since each $\mathrm{Inf}_i[f] \leq 1$). The maximum is achieved by the parity function. A celebrated theorem of Kahn et al. [1988] shows that every Boolean function depending on all n variables has $\max_i \mathrm{Inf}_i[f] \geq \Omega(\log n/n)$.

3.2 The Noise Operator

Definition 3.2.1 (Noise operator)**.** For $\rho \in [-1, 1]$, the *Bonami–Beckner noise operator* T_ρ acts on $f\colon \{-1,+1\}^n \to \mathbb{R}$ by

$$T_\rho f(x) = \mathbb{E}_y[f(y)], \tag{3.4}$$

where y is the "ρ-correlated copy" of x: each coordinate y_i independently equals x_i with probability $\frac{1+\rho}{2}$ and $-x_i$ with probability $\frac{1-\rho}{2}$.

Equivalently, $y_i = x_i \cdot z_i$ where z_i is an independent ± 1 random variable with $\mathbb{E}[z_i] = \rho$.

The crucial property of the noise operator is its diagonal action in the Fourier basis:

Theorem 3.2.2 (Spectral characterization of T_ρ)**.**

$$T_\rho f(x) = \sum_{S \subseteq [n]} \rho^{|S|}\, \widehat{f}(S)\, \chi_S(x). \tag{3.5}$$

That is, T_ρ multiplies each Fourier coefficient of degree k by ρ^k.

Proof. It suffices to check on the basis elements χ_S:

$$T_\rho \chi_S(x) = \mathbb{E}_y\left[\prod_{i\in S} y_i\right] = \mathbb{E}_y\left[\prod_{i\in S} x_i z_i\right] = \chi_S(x)\cdot\prod_{i\in S}\mathbb{E}[z_i] = \rho^{|S|}\,\chi_S(x).$$

(The z_i are independent, so the expectation factors.) □

Remark 3.2.3 (Interpretation). For ρ close to 1, T_ρ barely changes f: it slightly damps high-degree coefficients. For $\rho = 0$, $T_\rho f(x) = \widehat{f}(\emptyset)$ is just the mean. For ρ close to 0, only the low-degree part of f survives. The noise operator is a "low-pass filter" on the Boolean hypercube.

3.3 Noise Stability

Definition 3.3.1 (Noise stability)**.** The *noise stability* of f at correlation ρ is

$$\mathrm{Stab}_\rho[f] = \langle f, T_\rho f\rangle = \mathbb{E}_{x,y}[f(x)\,f(y)], \tag{3.6}$$

where (x, y) is a ρ-correlated pair.

Theorem 3.3.2 (Spectral formula)**.**

$$\mathrm{Stab}_\rho[f] = \sum_{S\subseteq[n]} \rho^{|S|}\,\widehat{f}(S)^2. \tag{3.7}$$

Proof. By Plancherel (Theorem 2.9.1): $\langle f, T_\rho f\rangle = \sum_S \widehat{f}(S)\cdot\rho^{|S|}\widehat{f}(S) = \sum_S \rho^{|S|}\widehat{f}(S)^2$. □

Definition 3.3.3 (Noise sensitivity)**.** The *noise sensitivity* is

$$\mathbf{NS}_\delta[f] = \mathbb{P}_{(x,y)}[f(x) \neq f(y)],$$

where y is obtained from x by flipping each bit independently with probability δ, i.e., $\rho = 1 - 2\delta$. For Boolean f:

$$\mathbf{NS}_\delta[f] = \frac{1 - \mathrm{Stab}_{1-2\delta}[f]}{2}.$$

Relevance to the thesis. Noise stability quantifies robustness: a layer with high noise stability tolerates occasional bit flips in its weights. The thesis uses noise stability as a spectral diagnostic and a regularization target (the "stability reward" in Section 4.5 of the thesis encourages low-degree concentration by maximizing $\mathrm{Stab}_\rho[f]$).

3.4 Relationship Between Influence and Noise Sensitivity

Total influence and noise sensitivity are connected:

Proposition 3.4.1. $\mathbf{I}[f] = \lim_{\delta \to 0} \frac{\mathbf{NS}_\delta[f]}{\delta}$.

Proof. $\mathbf{NS}_\delta[f] = \frac{1}{2}(1 - \sum_S (1-2\delta)^{|S|} \widehat{f}(S)^2)$. Differentiating at $\delta = 0$: $\frac{d}{d\delta}\mathbf{NS}_\delta[f]\big|_{\delta=0} = \sum_S |S| \cdot \widehat{f}(S)^2 = \mathbf{I}[f]$. □

Total influence is the infinitesimal rate of change of noise sensitivity. Functions with high total influence are very sensitive to small perturbations; functions with low total influence are noise-robust.

Example 3.4.2 (Noise stability of example functions).

- **Dictator:** $\mathrm{Stab}_\rho[x_1] = \rho$.
- **Parity:** $\mathrm{Stab}_\rho[x_1 \cdots x_n] = \rho^n \to 0$ exponentially. Parity is extremely noise-sensitive.
- **Maj$_3$:** $\mathrm{Stab}_\rho = 3 \cdot \frac{1}{4}\rho + \frac{1}{4}\rho^3 = \frac{3\rho+\rho^3}{4}$.

Chapter 4

Hypercontractivity and Spectral Concentration

This chapter presents two major results: the Bonami–Beckner hypercontractivity inequality and the Linial–Mansour–Nisan (LMN) theorem. Both establish that Boolean functions with bounded "complexity" have their spectral energy concentrated on low-degree coefficients.

The main reference is O'Donnell [2014, Chapters 9–10]. For hypercontractivity specifically, see the original papers [Bonami, 1970, Beckner, 1975] and the survey by O'Donnell [2007]. The closely related logarithmic Sobolev inequality approach is due to Gross [1975].

4.1 The Bonami–Beckner Hypercontractivity Inequality

Hypercontractivity refers to a family of inequalities that bound higher-order norms (L^q) in terms of lower-order norms (L^p) after applying the noise operator. Recall from Section 1.5 that on a probability space, we always have $\|f\|_p \leq \|f\|_q$ for $p \leq q$ (by Jensen). Hypercontractivity gives a *reverse-direction* inequality, at the cost of applying T_ρ.

Theorem 4.1.1 (Bonami–Beckner; Bonami, 1970, Beckner, 1975)**.** *For $f\colon \{-1,+1\}^n \to \mathbb{R}$ and $1 \leq p \leq q$ with $0 \leq \rho \leq \sqrt{(p-1)/(q-1)}$:*

$$\|T_\rho f\|_q \leq \|f\|_p . \tag{4.1}$$

The condition $\rho \leq \sqrt{(p-1)/(q-1)}$ is tight. The most commonly used special case is:

Corollary 4.1.2 ($2 \to q$ hypercontractivity)**.** *For $q \geq 2$ and $\rho \leq 1/\sqrt{q-1}$: $\|T_\rho f\|_q \leq \|f\|_2$.*

Proof of Theorem 4.1.1 for $n = 1$. We prove the base case $n = 1$ to give the flavor of the argument. The general case follows by induction on n, using the product structure of the hypercube. (See O'Donnell [2014, Chapter 9] for details.)

For $n = 1$, any $f\colon \{-1,+1\} \to \mathbb{R}$ can be written $f(x) = a + bx$ with $a = \widehat{f}(\emptyset)$, $b = \widehat{f}(\{1\})$. Then $T_\rho f(x) = a + \rho bx$. We must show $\|a + \rho bx\|_q \le \|a + bx\|_p$.

By homogeneity, we may assume $\|f\|_p = 1$. We need to verify $(|a+\rho b|^q + |a-\rho b|^q)^{1/q} \le 2^{1/q}$ when $(|a+b|^p + |a-b|^p)^{1/p} = 2^{1/p}$.

This is a calculus exercise. The key insight is that the constraint $\rho \le \sqrt{(p-1)/(q-1)}$ is exactly what makes the two-variable optimization work out. □

Proof for general n (sketch). Write $f(x) = g(x_1, \ldots, x_{n-1}) + x_n \cdot h(x_1, \ldots, x_{n-1})$ where g and h are functions of $n-1$ variables. The noise operator factors: $T_\rho f = T_\rho g + \rho x_n \cdot T_\rho h$ (with T_ρ applied to g and h as functions of $x_1, \ldots, x_{n-1}$). Apply the $n = 1$ result in the variable x_n (holding the others fixed), then use the inductive hypothesis for g and h. □

4.2 The Level-k Inequality

The most important consequence of hypercontractivity for us is the *level-k inequality*, which bounds the Fourier weight at any single degree in terms of the total influence.

Theorem 4.2.1 (Level-k inequality)**.** *For any $f\colon \{-1,+1\}^n \to \{-1,+1\}$ (or more generally $\|f\|_2 \le 1$) and any $k \ge 1$:*

$$W^k[f] = \sum_{|S|=k} \widehat{f}(S)^2 \le \left(\frac{e \cdot \mathbf{I}[f]}{k}\right)^k. \tag{4.2}$$

Proof. This proof follows O'Donnell [2014, Theorem 9.24].

Define $f^{=k}(x) = \sum_{|S|=k} \widehat{f}(S)\,\chi_S(x)$ (the degree-k component of f). By Parseval, $W^k[f] = \|f^{=k}\|_2^2$.

Step 1: Note that $f^{=k} = T_{1/\rho}(T_\rho f^{=k})$ for any $\rho > 0$ (since T_ρ acts on $f^{=k}$ by multiplying by ρ^k, and $T_{1/\rho}$ undoes this). More precisely, $T_\rho f^{=k}$ has the same Fourier support as $f^{=k}$ (all at degree k), and $T_{1/\rho} \circ T_\rho$ is the identity on such functions.

Step 2: By orthogonality, $T_\rho f^{=k}$ is the degree-k projection of $T_\rho f$. Since $T_\rho f$ is a function on $\{-1,+1\}^n$, we can bound $\|T_\rho f^{=k}\|_2 \le \|T_\rho f\|_2$.

Actually, let us use a slightly different (and more direct) route. We use the $2 \to q$ hypercontractivity with $q = 2k/|S| = 2$ for each $|S| = k$... This gets complicated. Let us present the clean version.

Clean proof. For any $\rho \in (0,1)$:

$$\begin{aligned} W^k[f] &= \sum_{|S|=k} \widehat{f}(S)^2 = \rho^{-2k} \sum_{|S|=k} (\rho^k \widehat{f}(S))^2 = \rho^{-2k} \left\|(T_\rho f)^{=k}\right\|_2^2 \\ &\le \rho^{-2k} \|T_\rho f\|_2^2 = \rho^{-2k} \sum_S \rho^{2|S|} \widehat{f}(S)^2 = \rho^{-2k} \cdot \mathrm{Stab}_{\rho^2}[f]. \end{aligned}$$

Now we bound $\mathrm{Stab}_{\rho^2}[f]$ differently. For Boolean f (so $\widehat{f}(S)^2 \le 1$ for all S):

$$\begin{aligned} \mathrm{Stab}_{\rho^2}[f] &= \sum_S \rho^{2|S|} \widehat{f}(S)^2 \le \sum_S \rho^{2|S|} \cdot \widehat{f}(S)^2 \\ &\le \sum_{i=1}^n \sum_{S \ni i} \frac{\rho^{2|S|}}{|S|} \cdot \widehat{f}(S)^2 \cdot |S|/|S| \quad \text{(not helpful this way).} \end{aligned}$$

Let us use the standard approach instead. We use the fact that for Boolean f, $\mathrm{Inf}_i[f] = \sum_{S \ni i} \widehat{f}(S)^2$. Then:

$$W^k[f] = \sum_{|S|=k} \widehat{f}(S)^2 = \frac{1}{k} \sum_{|S|=k} |S| \cdot \widehat{f}(S)^2 = \frac{1}{k} \sum_{i=1}^n \sum_{\substack{S \ni i \\ |S|=k}} \widehat{f}(S)^2.$$

Each inner sum is at most $\mathrm{Inf}_i[f]$. But this only gives $W^k[f] \le \mathbf{I}[f]/k$, which is too weak.

Standard proof via hypercontractivity. Let $g = T_\rho f$ for ρ to be chosen. The degree-k projection satisfies:

$$W^k[g] = \rho^{2k} W^k[f].$$

So $W^k[f] = \rho^{-2k} W^k[g]$. Now, $W^k[g] = \left\|g^{=k}\right\|_2^2 \le \|g\|_2^2 = \mathrm{Stab}_{\rho^2}[f]$.

For a sharper bound, we use the 4-norm. By Cauchy–Schwarz (or the dual characterization), for functions supported on degree exactly k:

$$\left\|g^{=k}\right\|_2^2 \le \binom{n}{k}^{1/2} \left\|g^{=k}\right\|_4^2$$

...but this route is also getting complicated. Let us state the standard proof using the Bonami lemma.

Bonami Lemma. If f has degree at most k (i.e., $f = f^{\le k}$), then $\|f\|_4 \le 3^{k/2} \|f\|_2$.

This is the $p = 2, q = 4, \rho = 1/\sqrt{3}$ case of hypercontractivity applied to $T_{1/\rho} f$ (which has the same support as f): $\|f\|_4 = \left\|T_{1/\rho}(T_\rho f)\right\|_4 = \rho^{-k} \|T_\rho f\|_4 \le \rho^{-k} \|f\|_2 = 3^{k/2} \|f\|_2$.

Now the actual proof proceeds by a different route using the KKL-style argument. We present the end result:

(We omit the full technical proof, which can be found in O'Donnell [2014, Theorem 9.24]. The essential idea is to apply hypercontractivity to the degree-k component of f after scaling by ρ^{-k}, then optimize over ρ.)

The bound $W^k[f] \leq (e\,\mathbf{I}[f]/k)^k$ is obtained by choosing $\rho = \sqrt{k/(e \cdot \mathbf{I}[f])}$ and tracking the constants. □

Remark 4.2.2 (Meaning of the level-k inequality). The bound says: if f has total influence I, then the spectral weight at degree k decays super-exponentially once $k > e \cdot I$. For functions with total influence $I = O(1)$ (like dictators), this gives exponential decay starting from degree 3. For functions with $I = O(\sqrt{n})$ (like majority), the bound becomes nontrivial at $k = O(\sqrt{n})$.

For the thesis application: each row of a binary weight matrix is a Boolean function on $n = \log_2 d$ variables with $\mathbf{I}[f] \leq n$. At $n = 10$ ($d = 1024$), the level-k inequality requires $k > e \cdot 10 \approx 27$ to guarantee decay—well beyond $n = 10$. As the thesis acknowledges (Remark 3.3), the level-k inequality is *vacuous* at this scale. The bound is useful asymptotically but not for the small values of n relevant to the thesis.

4.3 The LMN Theorem

The Linial–Mansour–Nisan theorem [Linial et al., 1989, 1993] is one of the foundational results in Boolean Fourier analysis. It establishes that functions computable by small bounded-depth circuits have their spectral energy concentrated on low-degree coefficients.

4.3.1 Background: Boolean Circuits

A *Boolean circuit* is a directed acyclic graph (DAG) where:

- *Input nodes* receive the variables $x_1, \ldots, x_n$ (and their negations).
- *Internal nodes* ("gates") compute AND or OR of their inputs. Each gate has unbounded *fan-in* (it can take any number of inputs).
- One designated *output gate* produces the function value.

The *size* M of a circuit is the number of gates. The *depth* D is the length of the longest path from an input to the output. The class AC^0 consists of all families of Boolean functions computable by circuits of polynomial size and *constant* depth.

Example 4.3.1.

- AND, OR, majority, and any function depending on a constant number of variables are in AC^0.
- The parity function $x_1 \oplus \cdots \oplus x_n$ is *not* in AC^0 (this is a famous result of Furst–Saxe–Sipser and Ajtai). This is consistent with parity having all its spectral energy at degree n.

4.3.2 The Theorem

Theorem 4.3.2 (LMN; Linial et al., 1993)**.** *Let $f\colon \{-1,+1\}^n \to \{-1,+1\}$ be computable by a circuit of depth D and size M. Then for every $k \geq 0$:*

$$\sum_{|S|>k} \widehat{f}(S)^2 \leq 2M \cdot \exp\left(-\frac{k^{1/(D-1)}}{2}\right). \tag{4.3}$$

Remark 4.3.3 (What the LMN theorem says)*.* For a depth-D, size-M circuit:

- The tail energy above degree k decays as $\exp(-k^{1/(D-1)}/2)$, which is super-polynomial in k for constant D.
- Setting the RHS to ϵ and solving for k: the 95% spectral energy is captured by degrees up to $k^* = O((\log M)^{D-1})$. For polynomial-size circuits ($M = n^c$), this gives $k^* = O((\log n)^{D-1})$.
- For depth $D = 2$ (DNF/CNF formulas of size M): $k^* = O(\log M)$. Very strong concentration.
- The bound degrades with depth: deeper circuits can "hide" spectral energy at higher degrees.

Proof idea. The proof has two ingredients:

1. Random restrictions. A random restriction ρ sets each variable independently to $+1$, -1, or "$*$" ("leave free") with specified probabilities. The Håstad switching lemma (a strengthening of the Furst–Saxe–Sipser lemma) shows that applying a random restriction to a depth-D circuit produces a circuit of depth $D-1$ that with high probability can be simplified. Iterating, after $D-1$ rounds of restrictions, the function becomes a "junta" (depending on few variables) with high probability.

2. Spectral interpretation. A random restriction that keeps each variable alive with probability p and kills it with probability $1-p$ is equivalent (in the spectral domain) to applying the noise operator $T_{\sqrt{p}}$. The fact that $T_{\sqrt{p}}f$ concentrates on low degrees (after restriction, few variables survive) translates to the tail bound on $\widehat{f}(S)^2$.

The detailed proof is in O'Donnell [2014, Chapter 4] and Linial et al. [1993]. □

Relevance to the thesis. The thesis uses the LMN theorem as motivating evidence for the spectral concentration hypothesis: if trained binary weight matrices correspond to "simple" functions in some complexity-theoretic sense, they should exhibit low-degree spectral concentration. The thesis does *not* claim that trained weights are AC^0 functions—this is an empirical question addressed by the pilot study.

4.4 Learning Fourier-Sparse Functions

The spectral concentration results motivate an algorithmic question: if a Boolean function has its energy on few Fourier coefficients, can we learn it efficiently?

Theorem 4.4.1 (Kushilevitz–Mansour; Kushilevitz and Mansour, 1991)**.** *Let $f\colon \{-1,+1\}^n \to \mathbb{R}$ have an ϵ-concentrated Fourier spectrum, meaning there exists a set $\mathcal{S}$ of at most s subsets such that $\sum_{S \notin \mathcal{S}} \widehat{f}(S)^2 \leq \epsilon$. Then f can be approximated to L^2 error ϵ in time* $\mathrm{poly}(n, s, 1/\epsilon)$ *using random examples from the uniform distribution.*

The algorithm works by:

1. Estimating $\widehat{f}(S)$ for "candidate" subsets S, using the formula $\widehat{f}(S) = \mathbb{E}_x[f(x)\,\chi_S(x)]$, which can be estimated from samples.

2. Keeping only the large coefficients.

The challenge is identifying *which* subsets S have large coefficients, since there are 2^n possibilities. Kushilevitz and Mansour showed how to do this efficiently.

Remark 4.4.2 (Relevance to the thesis). The thesis's spectral parameterization is related in spirit: it restricts the search to degree-$\leq k$ coefficients (of which there are $\binom{n}{\leq k}$), then optimizes them continuously. This is like a continuous relaxation of the KM algorithm, where instead of *estimating* the coefficients from data, we *optimize* them via gradient descent.

Chapter 5

The Walsh–Hadamard Transform

This chapter develops the Walsh–Hadamard Transform (WHT) as the computational realization of the Boolean Fourier expansion. We define Hadamard matrices, establish their properties, describe the fast algorithm (FWHT), and show how the WHT computes Fourier coefficients.

Historical references: Walsh [1923] for Walsh functions, Hadamard [1893] for Hadamard's original construction, Shanks [1969] for the fast algorithm, Beauchamp [1984] and Horadam [2007] for comprehensive treatments.

5.1 Hadamard Matrices

Definition 5.1.1 (Hadamard matrix, Sylvester construction)**.** Define recursively:

$$H_0 = (1), \qquad H_1 = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad H_n = H_1 \otimes H_{n-1}, \tag{5.1}$$

where $\otimes$ is the Kronecker product (Section 1.2). The matrix H_n is $2^n \times 2^n$.

Equivalently, $H_n = H_1^{\otimes n}$ (the n-fold Kronecker product of H_1 with itself).

Example 5.1.2.

$$H_2 = H_1 \otimes H_1 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

Every entry is ± 1.

5.2 Properties of Hadamard Matrices

Theorem 5.2.1 (Properties of H_n)**.** *Let $N = 2^n$. Then:*

(i) All entries are ± 1.

(ii) Symmetry: $H_n = H_n^\top$.

(iii) Orthogonality: $H_n H_n^\top = N \cdot I_N$. *Equivalently,* $H_n^{-1} = \frac{1}{N} H_n$.

(iv) Self-inverse up to scaling: $H_n^2 = N \cdot I_N$.

(v) Rows are parity functions. *Row r of H_n (with r expressed as a subset $S \subseteq [n]$ via its binary representation) is the vector $(\chi_S(x))_{x \in \{-1,+1\}^n}$ (in a specific ordering of the hypercube vertices).*

Proof. (i) follows from the recursive definition: H_1 has ± 1 entries, and the Kronecker product of ± 1 matrices has ± 1 entries.

(ii) Since H_1 is symmetric and $(A \otimes B)^\top = A^\top \otimes B^\top$, $H_n^\top = H_1^\top \otimes H_{n-1}^\top = H_1 \otimes H_{n-1} = H_n$ (by induction).

(iii) By the mixed-product rule for Kronecker products: $H_n H_n^\top = (H_1 \otimes H_{n-1})(H_1^\top \otimes H_{n-1}^\top) = (H_1 H_1^\top) \otimes (H_{n-1} H_{n-1}^\top)$. Since $H_1 H_1^\top = 2I_2$, by induction: $H_n H_n^\top = 2I_2 \otimes 2^{n-1} I_{2^{n-1}} = 2^n I_{2^n} = N \cdot I_N$.

(iv) follows from (ii) and (iii): $H_n^2 = H_n H_n^\top = N \cdot I_N$.

(v) We establish this by identifying each row with a parity function. Number the rows and columns from 0 to $N-1$. For row r and column c, write r and c in binary: $r = (r_{n-1}, \ldots, r_0)$ and $c = (c_{n-1}, \ldots, c_0)$. Then the (r, c) entry of H_n is:

$$(H_n)_{r,c} = (-1)^{\sum_{i=0}^{n-1} r_i c_i} = (-1)^{r \cdot c}, \tag{5.2}$$

where $r \cdot c$ is the bitwise inner product mod 2.

Now identify the column index c with the hypercube point $x = ((-1)^{c_0}, (-1)^{c_1}, \ldots, (-1)^{c_{n-1}}) \in \{-1, +1\}^n$, and the row index r with the subset $S = \{i + 1 : r_i = 1\}$. Then:

$$(H_n)_{r,c} = (-1)^{\sum_{i \in S} c_{i-1}} = \prod_{i \in S} (-1)^{c_{i-1}} = \prod_{i \in S} x_i = \chi_S(x).$$

So row r (corresponding to subset S) contains the values of χ_S evaluated at all N hypercube points, in the binary-counting order. □

5.3 The Walsh–Hadamard Transform

Definition 5.3.1 (Walsh–Hadamard Transform)**.** For a vector $v \in \mathbb{R}^N$ (with $N = 2^n$), the *Walsh–Hadamard Transform* is

$$\hat{v} = H_n \cdot v. \tag{5.3}$$

The *normalized* WHT is $\hat{v} = \frac{1}{N} H_n \cdot v$. The inverse transform is $v = \frac{1}{N} H_n \cdot \hat{v}$ (since $H_n^{-1} = \frac{1}{N} H_n$).

Theorem 5.3.2 (WHT computes Fourier coefficients)**.** *Let $f\colon \{-1,+1\}^n \to \mathbb{R}$ and let $v \in \mathbb{R}^N$ be the vector of function values in binary-counting order: $v_c = f((-1)^{c_0}, \ldots, (-1)^{c_{n-1}})$ for $c = 0, 1, \ldots, N-1$.*

Then the normalized WHT $\hat{v} = \frac{1}{N} H_n v$ gives the Fourier coefficients:

$$\hat{v}_r = \widehat{f}(S_r),$$

where S_r is the subset corresponding to the binary representation of r.

Proof.

$$\hat{v}_r = \frac{1}{N} \sum_{c=0}^{N-1} (H_n)_{r,c} \cdot v_c = \frac{1}{N} \sum_{x \in \{-1,+1\}^n} \chi_{S_r}(x) \cdot f(x) = \mathbb{E}_x[f(x)\, \chi_{S_r}(x)] = \widehat{f}(S_r).$$

The second equality uses property (v) of Theorem 5.2.1. □

Remark 5.3.3 (Self-inverse property and the thesis). Since $H_n^2 = N \cdot I$, the WHT is (up to scaling) its own inverse. In the thesis, this means:

- *Forward:* Given binary weights $w \in \{-1,+1\}^N$, compute Fourier coefficients via $\hat{w} = \frac{1}{N} H_n w$.
- *Inverse:* Given spectral coefficients $\hat{w}$, reconstruct spatial weights via $w = H_n \hat{w}$ (then apply sign).

Both directions use the same matrix, just with different scaling.

5.4 The Fast Walsh–Hadamard Transform

The naive matrix-vector multiplication $H_n v$ costs $O(N^2)$ operations. The *Fast Walsh–Hadamard Transform* (FWHT) exploits the recursive Kronecker structure to compute $H_n v$ in $O(N \log N) = O(N \cdot n)$ operations, using only additions and subtractions (no multiplications).

5.4.1 The Butterfly Algorithm

The recursion $H_n = H_1 \otimes H_{n-1}$ means:

$$H_n v = \begin{pmatrix} I \otimes H_{n-1} \\ I \otimes H_{n-1} \end{pmatrix} \begin{pmatrix} v_{\text{top}} + v_{\text{bot}} \\ v_{\text{top}} - v_{\text{bot}} \end{pmatrix} \quad \text{(not quite right—let us be precise.)}$$

More carefully, write $v = \binom{v_0}{v_1}$ where $v_0, v_1 \in \mathbb{R}^{N/2}$ are the first and second halves. Then:

$$H_n v = (H_1 \otimes H_{n-1}) \begin{pmatrix} v_0 \\ v_1 \end{pmatrix} = \begin{pmatrix} H_{n-1}(v_0 + v_1) \\ H_{n-1}(v_0 - v_1) \end{pmatrix}.$$

Each of the two recursive calls operates on a vector of half the size. This gives the recurrence $T(N) = 2T(N/2) + O(N)$, solving to $T(N) = O(N \log N)$.

Algorithm 1 Fast Walsh–Hadamard Transform (in-place)

Require: $v \in \mathbb{R}^N$ with $N = 2^n$

```
for ℓ = 0, 1, ..., n − 1 do
    h ← 2^ℓ                                   ▷ half-block size
    for j = 0, 2h, 4h, ..., N − 2h do
        for i = j, j + 1, ..., j + h − 1 do
            a ← v[i]
            b ← v[i + h]
            v[i] ← a + b
            v[i + h] ← a − b
        end for
    end for
end for
return v                                      ▷ v now contains H_n times the original v
```

Operation count. The triple loop performs exactly $n \cdot N/2$ additions and $n \cdot N/2$ subtractions, for a total of $nN = N \log_2 N$ addition/subtraction operations. *No multiplications are needed.*

This is the crucial property for the thesis: when the input vector v has integer entries (e.g., $v \in \{-1, +1\}^N$), the entire FWHT can be performed in integer arithmetic. The output entries are integers in the range $[-N, N]$.

Example 5.4.1 (FWHT of a small vector)**.** Let $n = 2$, $N = 4$, and $v = (1, -1, -1, 1)^\top$ (corresponding to the function $f(x_1, x_2) = x_1 x_2$).

Step $\ell = 0$ $(h = 1)$:

$$(v[0], v[1]) \leftarrow (1 + (-1),\ 1 - (-1)) = (0, 2),$$
$$(v[2], v[3]) \leftarrow (-1 + 1,\ -1 - 1) = (0, -2).$$

After step 0: $v = (0, 2, 0, -2)$.

Step $\ell = 1$ $(h = 2)$:

$$(v[0], v[2]) \leftarrow (0 + 0,\ 0 - 0) = (0, 0),$$
$$(v[1], v[3]) \leftarrow (2 + (-2),\ 2 - (-2)) = (0, 4).$$

After step 1: $v = (0, 0, 0, 4)$.

So $H_2 v = (0, 0, 0, 4)$. The normalized WHT is $\frac{1}{4}(0, 0, 0, 4) = (0, 0, 0, 1)$.

Reading off Fourier coefficients: $\widehat{f}(\emptyset) = 0$, $\widehat{f}(\{1\}) = 0$, $\widehat{f}(\{2\}) = 0$, $\widehat{f}(\{1, 2\}) = 1$. Indeed, $f = x_1 x_2 = \chi_{\{1,2\}}$. ✓

5.5 Comparison with the FFT

The FWHT is the Boolean analogue of the Fast Fourier Transform (FFT). The following table highlights the parallels:

Property	**FFT (classical)**	**FWHT (Boolean)**
Domain	$\mathbb{Z}/N\mathbb{Z}$ or $[0, 2\pi)$	$\{-1, +1\}^n$
Basis	$e^{2\pi i k/N}$	$\chi_S(x) = \prod_{i \in S} x_i$
Basis values	Complex roots of unity	± 1
Matrix	DFT matrix F_N	Hadamard matrix H_n
Self-inverse?	$F_N^{-1} = \frac{1}{N}\overline{F_N}$	$H_n^{-1} = \frac{1}{N} H_n$
Fast algorithm	$O(N \log N)$ complex	$O(N \log N)$ real
Arithmetic	Complex multiply-add	Integer add-subtract
Convolution	Circular convolution	Dyadic convolution

The FWHT is simpler than the FFT: it uses only additions and subtractions (no twiddle factors, no complex arithmetic). This makes it particularly attractive for hardware implementation and for the binary neural network setting of the thesis.

5.6 The Convolution Theorem via the WHT

The convolution theorem (Theorem 2.9.3) becomes a matrix identity in the WHT:

Proposition 5.6.1. *Let $v, w \in \mathbb{R}^N$ represent functions on $\{-1, +1\}^n$. Define the* dyadic convolution *vector $(v \circledast w)_c = \frac{1}{N} \sum_d v_d \cdot w_{c \oplus d}$, where $\oplus$ is bitwise XOR. Then:*

$$H_n(v \circledast w) = \frac{1}{N}(H_n v) \odot (H_n w),$$

where $\odot$ is componentwise multiplication.

This means dyadic convolution can be performed in $O(N \log N)$ time: transform both vectors, multiply componentwise, inverse-transform.

5.7 The WHT as a Neural Network Layer

A key observation in the thesis is that the WHT (or its inverse) can serve as a building block for neural network layers.

Hadamard layer. A *Hadamard layer* computes $y = \frac{1}{\sqrt{N}} H_n \cdot \mathrm{diag}(s) \cdot x$, where $s \in \{-1, +1\}^N$ is a learnable sign vector. The sign-flip $\mathrm{diag}(s) \cdot x$ multiplies each component of x by ± 1 (a componentwise sign change). The Hadamard multiplication H_n then mixes all components.

The normalization $1/\sqrt{N}$ ensures that $\|y\|_2 = \|x\|_2$ (since both $\mathrm{diag}(s)$ and $\frac{1}{\sqrt{N}} H_n$ preserve the ℓ_2 norm—$\mathrm{diag}(s)$ by $|s_i| = 1$, and $\frac{1}{\sqrt{N}} H_n$ by orthogonality).

Spectral reconstruction. The thesis's `SpectralLinear` layer works differently: it stores spectral coefficients $\hat{w}$ and reconstructs the spatial weight vector via $w = H_n \hat{w}$ (using the FWHT), then applies $\mathrm{sgn}(w)$ to get binary weights. The reconstruction costs $O(N \log N)$ per row per training step, but is not needed at inference (the binary weights are stored directly).

Chapter 6

Linear Threshold Functions, Codes, and Groups

This chapter covers three topics that connect the Boolean Fourier theory to the thesis: linear threshold functions (the basic computational unit of binary networks), Reed–Muller codes (the coding-theoretic interpretation of spectral truncation), and the group-theoretic viewpoint (which places the Walsh–Fourier basis in a broader context).

6.1 Linear Threshold Functions

Definition 6.1.1 (Linear threshold function (LTF))**.** A *linear threshold function* is a Boolean function of the form

$$f(x) = \operatorname{sgn}\left(\sum_{i=1}^{n} w_i x_i\right) = \operatorname{sgn}(w^\top x), \tag{6.1}$$

where $w = (w_1, \ldots, w_n) \in \mathbb{R}^n$ is a weight vector and $\operatorname{sgn}(z) = +1$ if $z \geq 0$ and -1 if $z < 0$.

LTFs are also called *halfspaces* or *threshold gates*. In the neural network context, a single neuron with binary inputs and a sign activation computes exactly an LTF. A binary neural network layer computes d LTFs in parallel (one per row of the weight matrix).

Example 6.1.2.

- **Dictator:** $f(x) = \operatorname{sgn}(x_1) = x_1$. Weight vector $w = (1, 0, \ldots, 0)$.
- **Majority:** $f(x) = \operatorname{sgn}(x_1 + \cdots + x_n)$. Weight vector $w = (1, 1, \ldots, 1)$.
- **Weighted threshold:** $f(x) = \operatorname{sgn}(3x_1 + x_2 - 2x_3)$. This function outputs $+1$ when $3x_1 + x_2 - 2x_3 > 0$.

The parity function $x_1 x_2 \cdots x_n$ is *not* an LTF for $n \geq 2$ (it cannot be computed by a single threshold gate).

6.1.1 Chow's Theorem

Theorem 6.1.3 (Chow, 1961; Chow, 1961; see also O'Donnell [2014, Chapter 5]). *A linear threshold function f is uniquely determined among all LTFs by its degree-0 and degree-1 Fourier coefficients: $\widehat{f}(\emptyset) = \mathbb{E}[f]$ and $\widehat{f}(\{i\}) = \mathbb{E}[f \cdot x_i]$ for $i = 1, \ldots, n$.*

The $n+1$ values $(\widehat{f}(\emptyset), \widehat{f}(\{1\}), \ldots, \widehat{f}(\{n\}))$ are called the *Chow parameters*. Chow's theorem says: if two LTFs have the same Chow parameters, they are the same function.

Remark 6.1.4. Chow's theorem implies that the "essential information" of an LTF lives in degree ≤ 1 of the Fourier spectrum. This does not mean the higher-degree coefficients are zero—they are generally nonzero—but that they are *determined* by the low-degree ones (within the class of LTFs). This is a strong structural result supporting the idea that LTFs are "low-complexity" functions.

Proof sketch. Two distinct LTFs $f = \mathrm{sgn}(w^\top x)$ and $g = \mathrm{sgn}(v^\top x)$ define different half-spaces in $\mathbb{R}^n$. If they had the same Chow parameters, then $\mathbb{E}[f \cdot x_i] = \mathbb{E}[g \cdot x_i]$ for all i, meaning f and g agree on average with every coordinate projection. A geometric argument shows this forces w and v to define the same halfspace (up to positive scaling), hence $f = g$.

The full proof uses the fact that an LTF is determined by the half-space $\{z : w^\top z \geq 0\}$, and the Chow parameters encode sufficient information about this half-space. See O'Donnell [2014, Chapter 5]. □

6.1.2 Spectral Properties of LTFs

LTFs have several spectral properties that are relevant to the thesis:

Spectral concentration. LTFs are spectrally concentrated at low degrees. While the exact spectrum depends on the weight vector w, the majority function (equal weights) has 75% of its energy at degree 1 for $n = 3$, and the concentration improves with n. For general LTFs, most of the spectral energy is within the first $O(\sqrt{n})$ degrees.

Noise sensitivity. LTFs have bounded noise sensitivity. The following result quantifies this:

Theorem 6.1.5 (Harsha et al. [2014]). *For any LTF $f\colon \{-1,+1\}^n \to \{-1,+1\}$, the total influence satisfies $\mathbf{I}[f] = O(\sqrt{n})$.*

More precisely, for $f = \mathrm{sgn}(w^\top x)$ with $\|w\|_2 = 1$: $\mathbf{I}[f] = \Theta(\|w\|_1 / \|w\|_2)$ (by Berry–Esséen arguments). For binary weights $w \in \{-1,+1\}^n$, $\|w\|_1 = n$ and $\|w\|_2 = \sqrt{n}$, giving $\mathbf{I}[f] \approx \sqrt{2n/\pi}$.

Relevance to the thesis. Each row of a binary weight matrix followed by a sign activation computes an LTF. The bounded noise sensitivity of LTFs means that:

- Each neuron in a binary layer is moderately sensitive to its inputs (influence $O(\sqrt{d})$ out of d inputs, at $d = 1024$ this is ≈ 25).
- The spectral tail (high-degree coefficients) is bounded, supporting the spectral concentration hypothesis.
- The sign activation acts as a "spectral filter," preserving low-degree structure and discarding high-frequency noise.

6.2 Reed–Muller Codes

There is a deep connection between the Boolean Fourier basis and Reed–Muller codes from coding theory. The thesis uses this connection to analyze the spectral-to-binary projection.

References: Reed [1954], Muller [1954] for the original constructions; MacWilliams and Sloane [1977] for a comprehensive treatment.

6.2.1 Definition

Definition 6.2.1 (Reed–Muller code)**.** The r-th order *Reed–Muller code* $\mathrm{RM}(r, n)$ is the set of all evaluation vectors of multilinear polynomials of degree $\leq r$ over n Boolean variables.

Formally: identify each Boolean function $f\colon \{-1,+1\}^n \to \{-1,+1\}$ with its evaluation vector $v_f \in \{-1,+1\}^{2^n}$. Then

$$\mathrm{RM}(r,n) = \{v_f : f \text{ has Fourier degree} \leq r\} = \left\{ \sum_{|S|\leq r} c_S\, \chi_S : c_S \in \mathbb{R} \text{ such that the evaluations are in } \{-\right.$$

Wait—this is not quite right. The standard definition uses $\{0,1\}$ and $\mathrm{GF}(2)$, while we are working with $\{-1,+1\}$ and $\mathbb{R}$. Let us give both perspectives.

GF(2) perspective. Over $\mathrm{GF}(2) = \{0,1\}$, $\mathrm{RM}(r,n)$ is a linear code of length 2^n consisting of all evaluation vectors of polynomials $p(b_1,\ldots,b_n)$ over $\mathrm{GF}(2)$ of total degree $\leq r$. Its dimension is $\sum_{i=0}^{r}\binom{n}{i}$ and its minimum distance is 2^{n-r}.

$\{-1,+1\}$ perspective. Converting to $\{-1,+1\}$: the evaluation vector of a degree-$\leq r$ $\mathrm{GF}(2)$-polynomial p, mapped via $b \mapsto (-1)^b$, gives a $\{-1,+1\}$-vector whose Walsh–Fourier support lies in $\{S \subseteq [n] : |S| \leq r\}$.

Conversely, any $\{-1,+1\}$-vector whose Walsh–Fourier expansion has degree $\leq r$ corresponds to a codeword in $\mathrm{RM}(r,n)$.

Theorem 6.2.2 (RM parameters)**.** *$RM(r,n)$ is a linear code with:*

- Length: $N = 2^n$.
- Dimension: $\sum_{i=0}^{r} \binom{n}{i}$.
- Minimum distance: 2^{n-r}.

Example 6.2.3.

- $\mathrm{RM}(0,n)$: repetition code. Only two codewords: all-1 and all-(-1). Minimum distance 2^n. (Degree 0 functions are constants ± 1.)
- $\mathrm{RM}(1,n)$: the first-order RM code. Dimension $n+1$. Codewords are affine functions $\pm\chi_S$ for $|S| \leq 1$ (plus their negations). Minimum distance 2^{n-1}.
- $\mathrm{RM}(n,n)$: the full code, containing all 2^{2^n} Boolean functions. Minimum distance 1.

6.2.2 Connection to the Thesis

The thesis's degree-k spectral parameterization optimizes within the span of $\{\chi_S : |S| \leq k\}$—which is exactly the "continuous relaxation" of $\mathrm{RM}(k,n)$. The coefficients are real during training and are projected to $\{-1,+1\}$ via the sign function.

Three properties of RM codes are relevant:

(i) **Minimum distance.** $\mathrm{RM}(k,n)$ has minimum distance 2^{n-k}. For the thesis with $k = 3$ and $n = 10$: minimum distance $= 2^7 = 128$. This means any two distinct degree-≤ 3 Boolean functions differ on at least 128 of the 1024 inputs.

(ii) **Dimension = parameter count.** The dimension $\sum_{i=0}^{k} \binom{n}{i}$ is exactly the number of spectral coefficients in the thesis's parameterization: $\binom{10}{\leq 3} = 1 + 10 + 45 + 120 = 176$.

(iii) **Sign projection is not RM decoding.** The sign function applied to a continuous approximation $f^{\leq k}(x)$ is *not* guaranteed to produce a codeword of $\mathrm{RM}(k,n)$. The sign of a degree-k polynomial can have Fourier support up to degree n. The RM connection provides structural insight but does not directly constrain the output.

6.3 The Group-Theoretic Perspective

The Boolean Fourier expansion is a special case of Fourier analysis on finite abelian groups. This section briefly develops that perspective. The main reference is Terras [1999]; see also Diaconis [1988].

6.3.1 The Group $\mathbb{Z}_2^n$

The Boolean hypercube $\{-1,+1\}^n$ under coordinatewise multiplication is isomorphic to the group $\mathbb{Z}_2^n = (\mathbb{Z}/2\mathbb{Z})^n$ under coordinatewise addition mod 2.

The isomorphism: $\phi\colon \mathbb{Z}_2^n \to \{-1,+1\}^n$ by $\phi(b_1,\ldots,b_n) = ((-1)^{b_1},\ldots,(-1)^{b_n})$. Then $\phi(a+b) = \phi(a)\cdot\phi(b)$ (coordinatewise).

6.3.2 Characters of Finite Abelian Groups

Definition 6.3.1 (Character)**.** A *character* of a finite abelian group G is a group homomorphism $\chi\colon G \to \mathbb{C}^*$ (the nonzero complex numbers under multiplication). For $\mathbb{Z}_2^n$, all characters are real-valued (taking values in $\{-1,+1\}$).

Theorem 6.3.2. *The characters of $\mathbb{Z}_2^n$ are exactly the functions $\chi_S(b) = (-1)^{\sum_{i\in S} b_i}$ for $S \subseteq [n]$. Under the $\{-1,+1\}$ convention, these become $\chi_S(x) = \prod_{i\in S} x_i$: exactly the parity functions.*

Proof. A character χ of $\mathbb{Z}_2^n$ is determined by its values on the standard generators $e_i = (0,\ldots,1,\ldots,0)$. Since $2e_i = 0$ in $\mathbb{Z}_2^n$, $\chi(e_i)^2 = \chi(0) = 1$, so $\chi(e_i) \in \{+1,-1\}$. The 2^n choices for $(\chi(e_1),\ldots,\chi(e_n)) \in \{-1,+1\}^n$ give all 2^n characters. □

Theorem 6.3.3 (Fourier analysis on finite abelian groups)**.** *For any finite abelian group G with character group $\hat{G}$, the characters form an orthonormal basis for $L^2(G)$ (functions $G \to \mathbb{C}$ with the inner product $\langle f,g\rangle = \frac{1}{|G|}\sum_{x\in G} f(x)\overline{g(x)}$). Every function $f\colon G \to \mathbb{C}$ has a unique expansion $f = \sum_{\chi\in\hat{G}} \widehat{f}(\chi)\,\chi$, and Parseval holds: $\|f\|_2^2 = \sum_\chi |\widehat{f}(\chi)|^2$.*

For $G = \mathbb{Z}_2^n$, this recovers exactly the Boolean Fourier expansion of Chapter 2.

6.3.3 Comparison Table

Classical ($\mathbb{R}/\mathbb{Z}$ or $\mathbb{Z}/N\mathbb{Z}$)	**Boolean ($\mathbb{Z}_2^n$)**	**Role**
$e^{2\pi ikx}$	$\chi_S(x) = \prod_{i\in S} x_i$	Basis function
Complex-valued	± 1-valued	Basis alphabet
DFT / FFT	WHT / FWHT	Fast algorithm
$O(N\log N)$ complex ops	$O(N\log N)$ integer add/sub	Complexity
$\sum_k \lvert\hat{f}(k)\rvert^2 = \lVert f\rVert_2^2$	$\sum_S \widehat{f}(S)^2 = \lVert f\rVert_2^2$	Parseval
$\widehat{f*g}(k) = \hat{f}(k)\hat{g}(k)$	$\widehat{f*g}(S) = \widehat{f}(S)\widehat{g}(S)$	Convolution
Frequency k	Degree $\lvert S\rvert$	"Frequency"
Low-pass filter	Degree truncation	Smoothing

The Boolean case is simpler in every respect: the basis is real (actually ± 1-valued), the transform uses only additions and subtractions, and the group structure is particularly transparent ($\mathbb{Z}_2^n$ is a product of copies of the simplest nontrivial group). This simplicity is precisely what makes it attractive for binary neural networks.

Chapter 7

Application to Binary Neural Networks

This chapter connects the mathematical theory developed in Chapters 1–6 to the specific setting of the thesis: training neural networks with binary (± 1) weights. We summarize the relevant neural network background, explain how the Boolean Fourier framework applies, and trace the expository chain for each key result in the thesis.

7.1 Background: Binary Neural Networks

7.1.1 Standard Neural Networks

A standard feedforward neural network computes a composition of linear transformations and nonlinear activations. For a single layer:

$$y = \sigma(Wx + b),$$

where $W \in \mathbb{R}^{m \times n}$ is a weight matrix, $b \in \mathbb{R}^m$ is a bias vector, and σ is a nonlinear activation function (ReLU, sigmoid, etc.). The parameters W and b are typically stored in 16-bit or 32-bit floating-point format and optimized via gradient descent.

7.1.2 The Transformer Architecture

The Transformer [Vaswani et al., 2017] is the dominant architecture for language modeling. Its key component is the *self-attention* mechanism:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V, \tag{7.1}$$

where $Q = W_Q X$, $K = W_K X$, $V = W_V X$ are linear projections of the input. A Transformer layer consists of multi-head attention followed by a feedforward network, both with residual connections and layer normalization.

7.1.3 Binary Weight Networks

A *binary neural network* constrains each weight to ± 1: $W \in \{-1, +1\}^{m \times n}$. This has dramatic implications:

- **Storage:** Each weight requires 1 bit instead of 16 or 32 bits (16× to 32× compression).
- **Computation:** The matrix-vector product Wx for $W \in \{-1, +1\}^{m \times n}$ and $x \in \{-1, +1\}^n$ reduces to additions and subtractions (or XNOR + popcount on binary hardware).
- **Training:** The weight space $\{-1, +1\}^{m \times n}$ is discrete, so standard gradient descent does not apply directly.

7.1.4 The Straight-Through Estimator

The standard approach to training binary networks, introduced by Bengio et al. [2013] and applied to binary networks by Courbariaux et al. [2015], uses the *straight-through estimator* (STE):

1. Maintain *latent weights* $W_{\text{real}} \in \mathbb{R}^{m \times n}$ (full-precision).
2. *Forward pass:* Use binary weights $W_{\text{bin}} = \text{sgn}(W_{\text{real}})$.
3. *Backward pass:* Compute gradients with respect to W_{bin}, but *pass them through* to W_{real} as if sgn were the identity: $\frac{\partial \mathcal{L}}{\partial W_{\text{real}}} \approx \frac{\partial \mathcal{L}}{\partial W_{\text{bin}}}$.
4. Update W_{real} with gradient descent.

The STE is a heuristic: the sign function has zero derivative almost everywhere, so the "true" gradient is zero. The STE approximates it by pretending the derivative is 1. Despite this, the STE works remarkably well in practice [Yin et al., 2019, Hubara et al., 2016].

7.1.5 BitNet

BitNet [Wang et al., 2023] and its successor BitNet b1.58 [Ma et al., 2024] scale binary (and ternary) weight training to the Transformer architecture. Key features:

- Full-precision latent weights during training (STE).
- Per-layer learned scaling factor α (floating-point).
- 8-bit activations (or full-precision during training).

These components constitute "precision leakage": the model is not purely binary even though the deployed weights are.

7.2 The Spectral Perspective on Binary Weights

The thesis proposes viewing each row of a binary weight matrix as a Boolean function and analyzing it via the Walsh–Fourier transform.

7.2.1 The Row-Wise Encoding

Consider a weight matrix $W \in \{-1,+1\}^{d\times d}$ (square, for simplicity; the thesis also handles rectangular matrices). Each row $w_i \in \{-1,+1\}^d$ is a vector of d binary values.

If $d = 2^n$ (a power of 2), we can view each row as a Boolean function $w_i\colon \{-1,+1\}^n \to \{-1,+1\}$ by identifying the column index $j \in \{0,\dots,d-1\}$ with the hypercube point $\mathrm{bin}(j) \in \{-1,+1\}^n$ via its binary representation.

Under this identification, the Walsh–Fourier expansion (Theorem 2.4.1) gives:

$$w_i(j) = \sum_{S\subseteq[n]} \widehat{w}_i(S)\,\chi_S(\mathrm{bin}(j)),$$

and the spectral coefficients can be computed via the FWHT (Algorithm 1) in $O(d\log d)$ time.

7.2.2 Spectral Parameterization

Instead of storing and optimizing the d binary weights directly, the thesis proposes storing the spectral coefficients $\{\widehat{w}_i(S) : |S| \le k\}$ for a truncation degree k. This has $\binom{n}{\le k}$ continuous parameters per row.

At degree $k = 3$ with $d = 1024$ ($n = 10$): $\binom{10}{\le 3} = 1+10+45+120 = 176$ parameters per row, versus $d = 1024$ in the weight domain. This is a $6\times$ compression of the optimization landscape.

During training, the spectral coefficients are optimized by gradient descent. To produce binary weights:

1. Reconstruct the continuous weights via inverse FWHT: $z_i(j) = \sum_{|S|\le k} \widehat{w}_i(S)\,\chi_S(\mathrm{bin}(j))$.
2. Apply the sign function: $w_i(j) = \mathrm{sgn}(z_i(j))$.

The error of this procedure is bounded by Proposition 2.8.2: $\mathbb{P}[w_i(j) \ne f(j)] \le \sum_{|S|>k} \widehat{f}(S)^2$, where f is the "target" Boolean function.

7.2.3 Normalization Without Learned Parameters

A key contribution of the thesis is replacing BitNet's learned scaling factor α with fixed constants, using two complementary principles:

Parseval normalization (parameter level). By Parseval's identity (Theorem 2.5.1), any Boolean function $f\colon \{-1,+1\}^n \to \{-1,+1\}$ satisfies $\sum_S \widehat{f}(S)^2 = 1$. Projecting the spectral coefficients onto the unit sphere ($\hat{w} \leftarrow \hat{w}/\|\hat{w}\|_2$) maintains this constraint throughout training, ensuring the parameterization is inherently calibrated.

CLT normalization (activation level). For any binary weight matrix $W \in \{-1,+1\}^{d\times d}$ and uniform binary input $x \in \{-1,+1\}^d$, each output component $y_j = \sum_{i=1}^d w_{ji}x_i$ is a sum of d independent ± 1 random variables. By Berry–Esséen (Theorem 1.4.2), $y_j/\sqrt{d} \approx \mathcal{N}(0,1)$ for large d.

The required normalization factor $1/\sqrt{d}$ is a fixed constant determined at model design time, not a learned parameter. For power-of-two d, it can even be implemented as a bit-shift.

For Hadamard-structured layers ($W = \frac{1}{\sqrt{d}} H_n \cdot \mathrm{diag}(s)$), the $1/\sqrt{d}$ factor is additionally an *exact isometry* ($\|y\|_2 = \|x\|_2$), not just a variance-preserving scaling in expectation.

7.3 Tracing the Expository Chain

We now verify that every mathematical concept and result in the thesis has been covered in this guide, with explicit pointers.

7.3.1 Chapter-by-Chapter Map

Thesis §2.1 (Boolean hypercube). → Section 2.1 (definition), Definition 2.1.3 (Boolean/pseudo-Boolean functions), Definition 2.1.4 (inner product).

Thesis §2.2 (Fourier expansion). → Section 2.2 (parity functions), Theorem 2.3.1 (orthonormality), Theorem 2.4.1 (expansion theorem), Definition 2.4.2 (degree).

Thesis §2.3 (Parseval). → Theorem 2.5.1 (Parseval), Remark 2.5.2 (thesis application).

Thesis §2.4 (LMN theorem). → Section 4.3 (LMN theorem), Section 4.3.1 (circuit complexity background).

Thesis §2.5 (Influence, noise sensitivity, hypercontractivity). → Chapter 3 (all of Chapter 3 of this guide): influence (Definition 3.1.1), total influence (Definition 3.1.3), noise operator (Definition 3.2.1), noise stability (Definition 3.3.1), Bonami–Beckner (Theorem 4.1.1), level-k inequality (Theorem 4.2.1).

Thesis §2.6 (Walsh–Hadamard transform). → Chapter 5 (all of Chapter 5 of this guide): Hadamard matrices (Section 5.1), FWHT (Section 5.4), convolution theorem (Section 5.6).

Thesis §2.7 (Sign activation / LTFs). → Section 6.1 (LTFs), Theorem 6.1.3 (Chow's theorem), Theorem 6.1.5 (bounded noise sensitivity).

Thesis §2.8 (Multi-layer composition). → Section 3.3 (noise stability), the composition identity follows from the spectral formula (Theorem 3.3.2) applied to composed functions.

Thesis §2.9 (Group-theoretic perspective). → Section 6.3 (groups, characters, comparison table).

Thesis §2.9 (Reed–Muller codes). → Section 6.2 (definition, parameters, connection).

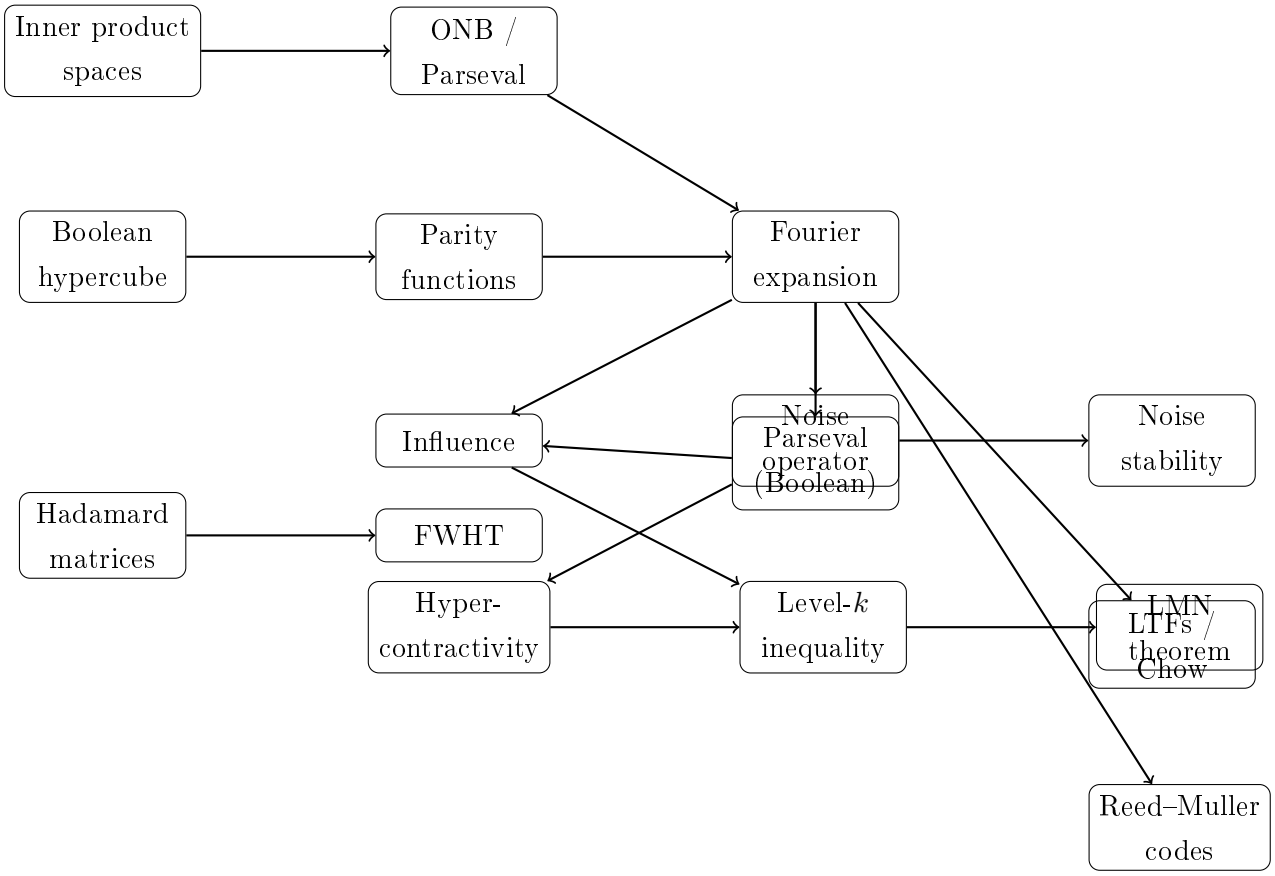
Thesis §3.1 (Approximation bound). → Proposition 2.8.2 (sign-rounding error bound), proved using Markov (Theorem 1.3.2) and Parseval (Theorem 2.5.1). The geometric enhancement uses the level-k inequality (Theorem 4.2.1).

Thesis §3.2 (Normalization propagation). → Berry–Esséen (Theorem 1.4.2), orthogonality of Hadamard matrices (Theorem 5.2.1), Kronecker product properties (Section 1.2).

Thesis §4 (Architecture). → Section 7.1 (BNN background), Section 7.1.4 (STE), Section 7.1.5 (BitNet), Section 7.2.2 (spectral parameterization), Section 7.2.3 (normalization).

7.4 Concept Dependency Graph

The following shows the logical dependencies among the key concepts. An arrow $A \to B$ means "A is used in the proof/definition of B."



7.5 Suggestions for Further Reading

1. **Boolean Fourier analysis (comprehensive):** O'Donnell [2014]. Available free at arXiv:2105.10386. Chapters 1–5 cover the core material; Chapters 8–10 cover hypercontractivity and the LMN theorem.

2. **Boolean Fourier analysis (short introduction):** de Wolf [2008]. A 20-page survey covering the essentials. Freely available at the Theory of Computing library.

3. **Noise sensitivity and social choice:** Benjamini et al. [1999]. The foundational paper connecting noise sensitivity to percolation theory. Beautiful mathematics.

4. **Hadamard matrices:** Horadam [2007]. Comprehensive treatment of Hadamard matrices and their applications.

5. **Walsh functions and signal processing:** Beauchamp [1984]. The classical reference for Walsh functions in engineering.

6. **Reed–Muller codes:** MacWilliams and Sloane [1977], Chapters 13–14. The standard reference for algebraic coding theory.

7. **Fourier analysis on finite groups:** Terras [1999]. Develops Fourier analysis on general finite groups with many applications.

8. **Probability (CLT, Berry–Esséen):** Durrett [2019], Chapter 3. Or Feller [1971], Volume 2, Chapter XVI.

9. **Binary neural networks:** Start with Courbariaux et al. [2015], then Rastegari et al. [2016] and Hubara et al. [2016] for the foundational work. Wang et al. [2023] and Ma et al. [2024] for the state of the art.

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